

Syllabus :- Sequence And Series

SECTION - I

Multiple Choice Questions (MCQ)

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- The sum of n terms of the series $1 + (1 + x) + (1 + x + x^2) + (1 + x + x^2 + x^3) + \dots$ where $x \neq 1$ is
 (1) $\frac{1}{1-x}$ (2) $\frac{n}{1-x}$ (3) $\frac{n}{1-x} - \frac{x(1-x^n)}{(1-x)^2}$ (4) n
- Let $a_1, a_2, a_3, \dots$ be a sequence of real numbers such that $|a_i| = |a_{i-1} + 1|$ for $i \geq 2$ and $a_1 = 0$. If A denotes the arithmetic mean of $a_1, a_2, \dots, a_n$ then which one of the following is correct?
 (1) $2nA = a_{2n+1} - n$ (2) $2nA = a_{2n} - n$ (3) $2nA = a_{2n+1} - n - 1$ (4) $2nA = a_{2n} - n - 1$
- If a, b, c are sides of a triangle then $(b + c - a)(c + a - b)(a + b - c) - abc$ is
 (1) positive (2) negative (3) non-positive (4) on-negative
- If a, x, b are in H.P. and a, y, z, b are in G.P., then the value of $\frac{yz}{x(y^3 + z^3)}$ is
 (1) ab (2) $\frac{1}{2ab}$ (3) $\frac{1}{2}ab$ (4) $2ab$
- If a, b, c are in A.P. with non-zero common difference and x, y, z are in G.P., then the point's (a, x) , (b, y) and (c, z) are collinear, if
 (1) $y^2 = z$ (2) $x = z^2$ (3) $x^2 = y$ (4) $x = y = z$
- Let a, b, c, d and e be real numbers such that $a > b > 0$ and $c > 0$. If $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P., b, c, d are in G.P. and c, d, e are in A.P., then $\frac{ab^2}{(2a-b)^2}$ is equal to :
 (1) c (2) $\sqrt{de}$ (3) e (4) d
- If sum of $n, 2n, 3n$ terms of an A.P. are S_1, S_2, S_3 respectively then the value of $\frac{S_3}{S_2 - S_1}$ is equal to
 (1) 4 (2) 5 (3) 2 (4) 3

8. In an arithmetic series of 16 terms with first term 16, the sum is equal to the square of the last term. The common difference of the series may be
- (1) $\frac{8}{5}$ (2) $-\frac{4}{5}$ (3) 0 (4) $\frac{2}{5}$
9. If the 3rd, 6th and last term of H.P. are $\frac{1}{3}, \frac{1}{5}, \frac{3}{203}$ then the number of terms in H.P. are
- (1) 100 (2) 150 (3) 200 (4) 250
10. Sum of infinite number of terms of a G.P. is 20 and sum of their square is 100. The common ratio of the G.P. is-
- (1) $\frac{1}{5}$ (2) $\frac{8}{5}$ (3) $\frac{3}{5}$ (4) 5
11. If 8 AMs are placed 2 and 17; then 5th AM is –
- (1) $6\frac{1}{3}$ (2) $7\frac{6}{3}$ (3) $31\frac{1}{3}$ (4) None
12. If the sum of n terms of two Arithmetic series are in the ratio $(7n + 1) : (4n + 27)$, then their 11th terms are in the ratio –
- (1) $\frac{3}{4}$ (2) $\frac{4}{3}$ (3) $\frac{78}{61}$ (4) None
13. If A, B, C are acute positive angles such that $A + B + C = \pi$ and $\cot A \cot B \cot C = k$ then
- (1) $k \leq \frac{1}{3\sqrt{3}}$ (2) $k \geq \frac{1}{3\sqrt{3}}$ (3) $k < \frac{1}{9}$ (4) $k > \frac{1}{3}$
14. If a, b, c are in A.P., $a + b + c = 60$, $(a - 2)$, b, $(c + 3)$ are in G.P., then possible values of $a^2 + b^2 + c^2$ are
- (1) 1218 (2) 1208 (3) 1288 (4) 1296
15. The sum of 3 numbers in GP is 42. If first two numbers are increased by 2 and third is decreases by 4 then resulting numbers form an A.P. Then the numbers of G.P. are -
- (1) 3, 6, 12 (2) 6, 12, 24 (3) 4, 8, 16 (4) None
16. If $\log_2(a + b) + \log_2(c + d) \geq 4$. Then the min. value of the expression $a + b + c + d$ is -
- (1) 2 (2) 4 (3) 8 (4) None of these
17. If reciprocals of the roots of equation $10x^3 - cx^2 - 54x - 27 = 0$ are in arithmetic progression, then find the value of c
- (1) 5 (2) 6 (3) 8 (4) 9
18. If p and q are positive real numbers such that $p^2 + q^2 = 1$, then find the maximum value of p + q
- (1) 1.5 (2) 1.42 (3) 1.40 (4) 1.41

19. If a, b, c are positive numbers and $ab^2c^3 = 64$ then find least value of $\left(\frac{1}{a} + \frac{2}{b} + \frac{3}{c}\right)$
- (1) 2 (2) 3 (3) 1 (4) 4
20. If the sum of the first n terms of the series $\sqrt{3} + \sqrt{75} + \sqrt{243} + \sqrt{507} + \dots$ is $435\sqrt{3}$, then find n
- (1) 12 (2) 14 (3) 18 (4) 15

SECTION-II

(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Nine (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. Let $A_n = \left(\frac{3}{4}\right) - \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 - \dots + (-1)^{n-1} \left(\frac{3}{4}\right)^n$ and $B_n = 1 - A_n$. Then find the least odd natural number p, so that $B_n > A_n$, for all $n \geq p$
22. If $x_1, x_2, \dots, x_n$ and $\frac{1}{h_1}, \frac{1}{h_2}, \dots, \frac{1}{h_n}$ are two A.P. such that $x_3 = h_2 = 8$ and $x_8 = h_7 = 20$, then find value of $\frac{x_5 h_{10}}{100}$
23. If $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ are two A.P.s, with common difference α, β respectively and $a_1 b_1 = 120$, $\alpha \beta = -6$, $a_1 \beta + b_1 \alpha = 29$, then $a_7 \cdot b_7$ is equal to
24. A tree, in each year, grows 5 cm less than it grew in the previous year. If it grew half a metre in the first year, then find the height of the tree (in metres), when it ceases to grow
25. Let a_1, a_2, a_3, a_4, a_5 be a G.P. of positive real numbers such that the A.M. of a_2 and a_4 is 117 and the G.M. of a_2 and a_4 is 108. If the A.M. of a_1 and a_5 is A then find $\frac{A}{10}$

Answer Key_DPP-05

1.	(3)	2.	(1)	3.	(3)	4.	(2)	5.	(4)	6.	(3)	7.	(4)
8.	(3)	9.	(1)	10.	(3)	11.	(3)	12.	(2)	13.	(1)	14.	(2)
15.	(2)	16.	(3)	17.	(4)	18.	(4)	19.	(2)	20.	(4)	21.	(7)
22.	(25.60)	23.	(78)	24.	(2.75)	25.	(14.55)						